Python + MYSQL Mini-Project

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Vehicle Rental System

Project Overview
The Vehicle Rental Management System is a Python and MySQL-based console application developed to manage vehicle rentals efficiently. The system allows administrators to manage vehicle and customers, rent vehicle, process returns, and maintain rental history.
This project demonstrates practical implementation of database connectivity, CRUD operations, business logic, and menu-driven programming using Python and MySQL.

Tools & Technologies Used

- Python 3.14.3
- MySQL 8.0 Database
- MySQL X DevAPI (mysqlx)
- VS-Code

Database Design

Vehicle Table
Stores vehicle’s information.
Fields:

- Vehicle ID
- Vehicle Name
- Registration Number
- Rent Per Day
- Fuel Type
- Category (5/7/13 Seater)
- Status (Available/Rented/Maintenance)

```
mysql> desc vehicle;
+-----------+-------------+------+-----+-----------+----------------+
| Field     | Type        | Null | Key | Default   | Extra          |
+-----------+-------------+------+-----+-----------+----------------+
| v_id      | int         | NO   | PRI | NULL      | auto_increment |
| v_name    | varchar(20) | YES  |     | NULL      |                |
| reg_no    | varchar(20) | YES  | UNI | NULL      |                |
| v_rent    | int         | YES  |     | NULL      |                |
| fuel_type | varchar(20) | YES  |     | NULL      |                |
| category  | varchar(5)  | YES  |     | NULL      |                |
| status    | varchar(15) | YES  |     | available |                |
+-----------+-------------+------+-----+-----------+----------------+
7 rows in set (0.10 sec)
```

Database Design

Customer Table
Stores customer's details.
Fields:

- Customer ID
- Customer Name
- Contact Number
- Address

```
mysql> desc customer;
+--------------+-------------+------+-----+---------+----------------+
| Field        | Type        | Null | Key | Default | Extra          |
+--------------+-------------+------+-----+---------+----------------+
| c_id         | int         | NO   | PRI | NULL    | auto_increment |
| c_name       | varchar(20) | YES  |     | NULL    |                |
| c_contact_no | varchar(15) | YES  | UNI | NULL    |                |
| c_address    | varchar(30) | YES  |     | NULL    |                |
+--------------+-------------+------+-----+---------+----------------+
4 rows in set (0.01 sec)
```

Database Design

Rent Table
Stores rental transactions.
Fields:

- Rent ID
- Vehicle ID
- Customer ID
- Rent Date
- Rent Days
- Return Date
- Total Amount
- Rent Status

```
mysql> desc rent;
+--------------+---------------+------+-----+-----------+-------------------+
| Field        | Type          | Null | Key | Default   | Extra             |
+--------------+---------------+------+-----+-----------+-------------------+
| r_id         | int           | NO   | PRI | NULL      | auto_increment    |
| v_id         | int           | NO   | MUL | NULL      |                   |
| c_id         | int           | NO   | MUL | NULL      |                   |
| r_date       | date          | YES  |     | curdate() | DEFAULT_GENERATED |
| rent_days    | int           | NO   |     | NULL      |                   |
| total_amount | decimal(10,2) | YES  |     | NULL      |                   |
| return_date  | date          | YES  |     | NULL      |                   |
| rent_status  | varchar(15)   | YES  |     | active    |                   |
+--------------+---------------+------+-----+-----------+-------------------+
8 rows in set (0.00 sec)
```

System Preview

Main Menu

```
===== VEHICLE RENTAL SYSTEM =====

1. Vehicle
2. Customer
3. Rent Vehicle
4. Return Vehicle
5. Rental History
6. Exit
Enter Choice: 
```

The application starts with a simple menu-driven interface that allows users to navigate through different modules, including Vehicle Management, Customer Management, Vehicle Rental, Vehicle Return, and Rental History.

Vehicle Management Module

```
Enter Choice: 1

===== VEHICLE MENU =====
1. Add Vehicle
2. View Vehicles
3. Change Vehicle Status
4. Delete Vehicle
5. Back
Enter Choice: 
```

The Vehicle Management module allows the administrator to manage the vehicle inventory. It provides options to add new vehicles, view all registered vehicles, update the availability status of a vehicle, and delete vehicle records when required.

Customer Management Module

```
===== CUSTOMER MENU =====
1. Add Customer
2. View Customers
3. Delete Customer
4. Back
Enter Choice: 
```

The Customer Management module is responsible for maintaining customer information required for vehicle rental operations. It allows the administrator to register new customers, view customer records, and remove customer information when necessary.

Rent Vehicle Module

```
Enter Choice: 3

 AVAILABLE VEHICLES
-----------------------------------------------------
1006    Toyota Innova Crysta    MH12BX2503      3000

 CUSTOMERS
106     Amit Pande
107     Karan Shinde

Enter Customer ID: 107
Enter Vehicle ID: 1006
Enter Rent Days: 3

===== RENT SUMMARY =====
Vehicle : Toyota Innova Crysta
Rent Days : 3
Return Date : 2026-07-08
Total Amount : 9000

Confirm Rent (Y/N): y

 Vehicle Rented Successfully!
Happy Journey 😃
```

The Vehicle Rental module enables customers to rent available vehicles. It displays only vehicles with an Available status, ensuring that rented or unavailable vehicles cannot be selected. After the rental is confirmed, the system records the transaction and updates the vehicle's status automatically.

Vehicle Return Module

```
1. Vehicle
2. Customer
3. Rent Vehicle
4. Return Vehicle
5. Rental History
6. Exit
Enter Choice: 4

---- RENT RECORDS ----
Rent ID Vehicle ID      Customer ID     Status
--------------------------------------------------
9       1004            106             active
10      1006            107             active
Enter Rent ID: 9
Vehicle Returned Successfully!

1. Vehicle
2. Customer
3. Rent Vehicle
4. Return Vehicle
5. Rental History
6. Exit
```

The Vehicle Return module manages the return process for rented vehicles. It displays all active rental records, allowing the administrator to select the appropriate Rent ID. Once the return is confirmed, the rental status and vehicle availability are updated automatically.

Rental History Module

```
1. Vehicle
2. Customer
3. Rent Vehicle
4. Return Vehicle
5. Rental History
6. Exit
Enter Choice: 5
------------------------------------------------------------------------------------------------------
RID     CUSTOMER        VEHICLE REGNO   DAYS    RETURN DATE     AMOUNT  STATUS
------------------------------------------------------------------------------------------------------
11      Amit Pande      Hyundai i10     MH12AB1234      3       2026-07-08 00:00:00     4500.00 active
```

The Rental History module provides a complete overview of all rental transactions. It helps administrators track previous and ongoing rentals by displaying customer information, vehicle details, rental duration, total amount, and rental status.

Conclusion

Through this project, I strengthened my understanding of Python programming, relational database design, SQL, foreign key relationships, and real-world application development. It also enhanced my ability to build modular, database-driven software solutions.